SAR Jamming Recognition via Discriminative Feature Distance Metrics Under Imbalanced Sample

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***Abstract*— Accurately recognizing the type of complex electromagnetic jamming is the essential prerequisite for synthetic aperture radar (SAR) anti-jamming. However, current convolutional neural network (CNN)-based SAR jamming recognition methods require balanced training samples, which contradicts the varying difficulty of acquiring various jamming types, drastically reducing the recognition accuracy and generalization ability. This article proposes a discriminative feature distance metric model, JRSNet, for jamming recognition under imbalanced training samples, by refining the jamming modulation differences in the time–frequency (TF) domain into discriminative features. Novel feature discriminative distance metric (FD^2M) loss function and discriminative feature constraint module (DFCM) are put forward to guarantee JRSNet learns embedding expression paradigm from jamming TF spectrograms to discriminative features, thus eliminating the influence of imbalanced training samples. Moreover, new spatial and channel attention modules are incorporated into JRSNet to capture jamming modulation information from multiple dimensions, consequently further improving recognition accuracy. Precisely because of the captured modulation regions in feature maps by spatial attention, the proposed approach can achieve jamming suppression synchronously. Experimental results show that under imbalanced training samples, JRSNet can accurately identify multiple jamming types both within and outside the training dataset with high generalizability. Compared with the existing jamming recognition methods, JRSNet performs superior recognition while taking into account good jamming suppression performance.**

***Index Terms*— Feature distance metric, imbalanced training sample, jamming recognition, jamming suppression, synthetic aperture radar (SAR).**

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I. Introduction

SYNTHETIC aperture radar (SAR), an advanced microwave imaging device, is a crucial tool for strategic intelligence and battlefield reconnaissance since it has high-resolution imaging ability in all-day and all-weather [1], [2], [3]. However, as electronic jamming technology advances rapidly, jamming with complex modulations and varying parameters [4], [5], severely corrupts SAR signals and impedes SAR image interpretation [6], thus threatening the survival and combat of SAR [7], [8]. To address this issue, it is urgent to develop SAR anti-jamming technologies [9]. The known types of SAR jamming are prerequisites for effective anti-jamming in most scenarios [10], [11], [12]. Therefore, how to accurately identify jamming types has attracted wide attention in the field of SAR electronic countermeasures [13], [14].

Obtaining as sufficient and effective features as possible and designing suitable rules are the keys for SAR to recognize different jamming signals accurately [15]. Traditional jamming recognition algorithms manually extracted features in the time domain [16], frequency domain [17], [18], [19], time–frequency (TF) domain [20] and image domain [21], and then designed classifiers/thresholds [such as support vector machine (SVM)] to recognize jamming. However, selecting features manually is time-consuming and relies on the expert prior, which leads to unstable feature quality, and cannot guarantee the jamming recognition performance [14]. Overall, these algorithms perform poor robustness and only apply to limited jamming types.

Recently, deep learning has been favored by SAR signal processing tasks for its remarkable nonlinear fitting and automatic feature extraction capabilities [22], [23], [24], [25], [26], [27]. Inspired by these successful applications, some researchers have developed jamming recognition based on deep learning. [13] constructed a convolutional neural network (CNN) to mine power spectrum characteristics of jamming signals for recognition. In [28], 1D-CNN and S-CNN were designed to recognize ten jamming signals under sufficient training samples, after fusing real and imaginary parts of time domain jamming signals. By analyzing the jamming location and texture differences in SAR images, [29] proposed an attention-based VGG16 (VGG16-A) model for jamming recognition. In [14], weighted ensemble learning was introduced to

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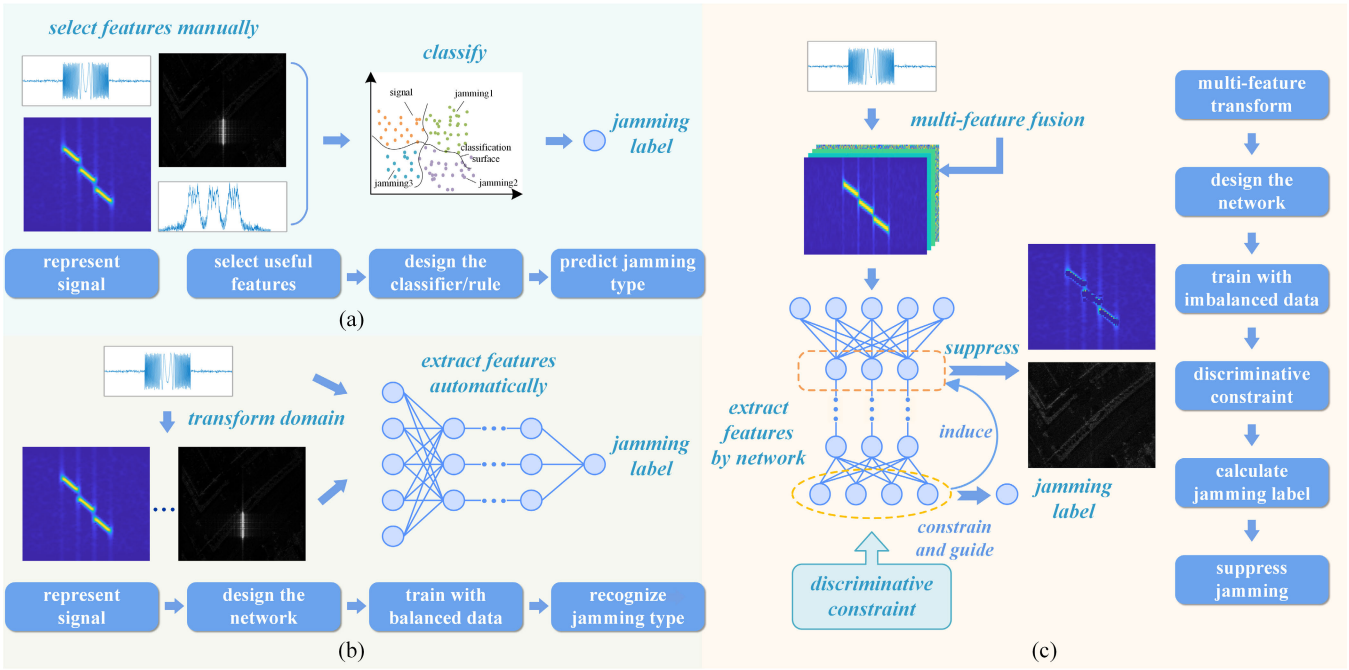


Fig. 1. Comparison of different jamming recognition methods. (a) and (b) Basic ideas of the existing two categories of SAR jamming recognition methods, respectively. The traditional methods in (a) predict jamming types by manually selecting features and designing classifiers or rules. The existing CNN-based methods in (b) used balanced jamming samples to train CNN to identify jamming types. (c) Proposed framework based on MFF guides the JRSNet by discriminative constraint to obtain discriminative features, and then calculates jamming types and suppresses jamming signals.

CNN (WECNN) to improve jamming recognition accuracy and generalization. Based on contrastive learning, [30] proposed an aggregated-attention model in the TF domain, which captured long-range spatial contextual information, thus identifying accurately multiple jamming. Luo et al. [31] designed a knowledge distillation jamming recognition network (JR-TFSAD) to shorten inference time while guaranteeing recognition performance.

Although the aforementioned deep learning-based methods can accurately recognize multiple jamming, they usually assume balanced jamming samples and complete jamming types in the training set. However, the noncooperative nature and confidentiality of real jamming scenes prevent the available jamming data from satisfying the above assumptions. The inaccessible new-type jamming also exacerbates the sample imbalance. In such cases, these methods often tend to have poor recognition performance. In addition, most methods compel models to map different jamming to an independent probability interval [0, 1] for recognition, ignoring commonality and intrinsic modulation differences between various jamming, which results in inadequate generalization under imbalanced training samples.

In this article, to tackle imbalanced training samples, based on the physical characteristic that various modulations of the same signal generate different jamming, we propose a novel jamming recognition idea via discriminative feature distance metrics, as shown in Fig. 1. Then, a specific model, JRSNet, is derived from this idea. JRSNet captures these modulation differences in jamming TF spectrograms and learns to feature embedding expression paradigm, thus extracting the discriminative features of jamming. The jamming type is recognized by the nearest distance metric between the discriminative features of the jamming sample and query samples. In this way, avoiding limiting the numbers of each jamming type, accurate and generalized jamming recognition under imbalanced training samples can be solved naturally. The main contributions of this article are as follows.

1) We propose a new scheme, JRSNet, to recognize jamming under imbalanced training samples via discriminative feature distance metric. Among JRSNet, dense block attention (DBA) and channel aggregate attention (CAA) are designed to refine discriminative information from multiple dimensions of jamming TF spectrograms.
2) To eliminate the recognition performance degradation caused by imbalanced training samples, the feature discriminative distance metric (FD^2M) loss and the discriminative feature constraint module (DFCM) are put forward. They constrain and induce JRSNet to learn the embedding expression paradigm from jamming TF spectrograms to discriminative features.
3) Numerical experiments and visualization results demonstrate that under imbalanced training samples, JRSNet can recognize numerous jamming types within/outside the training dataset with high accuracy and generalization.
4) Considering the captured jamming modulation differences are also exactly the key to jamming suppression, this work extends JRSNet to suppress jamming signals in the TF spectrogram through the outputs of DBA. This is an exploration of jamming joint recognition and suppression, which undoubtedly reduces resource consumption.

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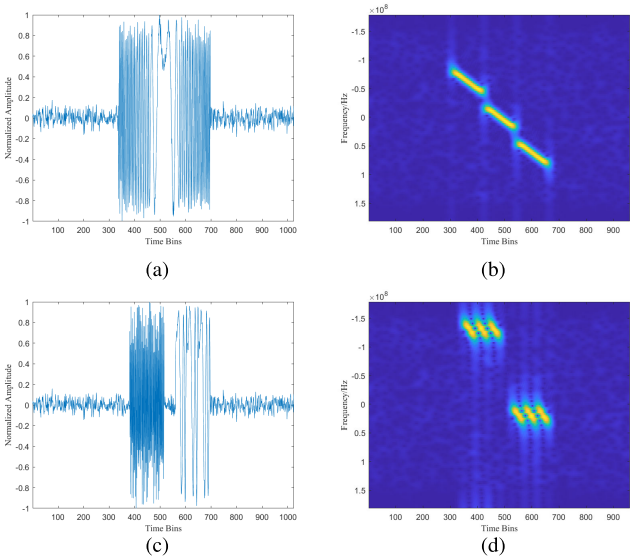


Fig. 2. Time domain waveforms of (a) SSFJ and (c) ISRJ, and TF spectrograms of (b) SSFJ and (d) ISRJ.

The rest of this article is organized as follows. Section II briefly describes the generation of the SAR jamming data used for the experiments. Section III analyzes the proposed SAR jamming recognition mechanism and formulation with jamming suppression. Section IV describes in detail the architecture of the proposed JRSNet. Section V presents the experimental details, and comprehensively analyzes the jamming recognition, suppression, and ablation results. Finally, Section VI discusses and concludes the whole article.

II. SAR Jamming Data Generation

This section first analyzes the multidomain characteristics of SAR jamming signals. Then, the generation process of the jamming data used in our work is presented.

A. TF Analysis

For an interfered SAR system, the received pulse echo $x(k)$ consists of the target signal $s(k)$, jamming $j(k)$ and background noise $n(k)$, which can be formulated as follows:

$$x(k) = s(k) + j(k) + n(k) \tag{1}$$

where k is the kth range time sample. Fig. 2(a) and (c) shows the time domain waveforms of subsection shift frequency jamming (SSFJ) and interrupted sampling and repeating jamming (ISRJ) when the signal-to-noise ratio (SNR) is 0. There are modulation differences in time and frequency between these two jamming signals. Hence, the short-time Fourier transform (STFT) [12] is employed to track frequency distribution over time

$$\begin{aligned} \boldsymbol{X}(t, f) &= \sum_k x(k)h(k-t)e^{-j\frac{2\pi f}{M}k} \\ &= \boldsymbol{S}(t, f) + \boldsymbol{J}(t, f) + \boldsymbol{N}(t, f) \end{aligned} \tag{2}$$

where f is the frequency sampling point, $h(k-t)$ represents the sliding window, M is the window length, $\boldsymbol{X}$, $\boldsymbol{S}$, $\boldsymbol{J}$, and $\boldsymbol{N}$ are the TF spectrograms of the corresponding four items in (1), respectively. Fig. 2(b) and (d) is TF spectrograms of SSFJ and ISRJ. The modulation differences of jamming signals in the TF domain are clearer, more clustered, and more typical, which facilitates jamming recognition.

B. Multifeature Tensor (MFT) Data Generation

The prerequisite for capturing jamming modulation differences is adequate retention of useful information in TF spectrograms. Since TF spectrograms are complex-valued, multifeature fusion (MFF) is introduced to process them. Fig. 3 is flowchart of MFF, $\boldsymbol{X}_M$, $\boldsymbol{X}_R$, $\boldsymbol{X}_I$, and $\boldsymbol{X}_P$ represent the modulus, real part, imaginary part, and phase of $\boldsymbol{X} \in \mathbb{C}^{t\times f}$, respectively. The MFF module fusion rules are as follows:

$$\begin{aligned} \boldsymbol{Y} &\triangleq \mathcal{T}_{\mathrm{MF}}(\boldsymbol{X}) \\ &\triangleq \mathcal{T}_{\mathrm{MF}}(\boldsymbol{X}_M, \boldsymbol{X}_R, \boldsymbol{X}_I, \boldsymbol{X}_P) \\ &\triangleq [\alpha_M \boldsymbol{X}_M; \alpha_R \boldsymbol{X}_R; \alpha_I \boldsymbol{X}_I; \alpha_P \boldsymbol{X}_P] \end{aligned} \tag{3}$$

where

$$\alpha_M, \alpha_R, \alpha_I, \alpha_P \in [0, 1], \quad \boldsymbol{Y} \in \mathbb{R}^{t\times f\times 4} \tag{4}$$

where $\mathcal{T}_{\mathrm{MF}}$ is MFF, $[\cdot]$ represents matrix concat, and α_M, α_R, α_I, and α_P are scale factors. As seen from Fig. 3, jamming modulation differences are more noticeable in $\boldsymbol{X}_M$, $\boldsymbol{X}_R$, and $\boldsymbol{X}_I$. So, we set α_M, α_R, and α_I to 1 and α_P to 0.8. $\boldsymbol{Y}$ is the MFT generated by MFF, which fully retains the jamming modulation information. Thus, the MFT is used as the input to the jamming recognition network, facilitating its accurate capture of jamming modulation difference features.

III. SAR Jamming Recognition Mechanism and Formulation

This section details the proposed feature distance metric SAR jamming recognition mechanism and formulation to overcome imbalanced training samples. After that, the principle of extending this recognition mechanism to jamming suppression is explained.

A. Proposed Feature Distance Metric Jamming Recognition Mechanism and Formulation

For most of SAR jamming signals, they can be represented as follows:

$$j(k) = \sum_n A_n \cdot s(k - \tau_n(k)) \cdot \exp(\varphi_n(k)) \tag{5}$$

where A_n, $\varphi_n(k)$, and $\tau_n(k)$ are the amplitude, frequency modulation, and time delay of nth jamming signal, respectively. Obviously, different types of jamming signals can be regarded as different modulations of the target signal $s(k)$ in terms of frequency, amplitude, and phase, such as the clear frequency modulation difference between SSFJ and ISRJ in the TF spectrogram at Fig. 2. These intrinsic modulation differences exactly imply the key to jamming recognition. Theoretically, jamming can be recognized by capturing and refining these differences from TF spectrograms using CNN, which does not

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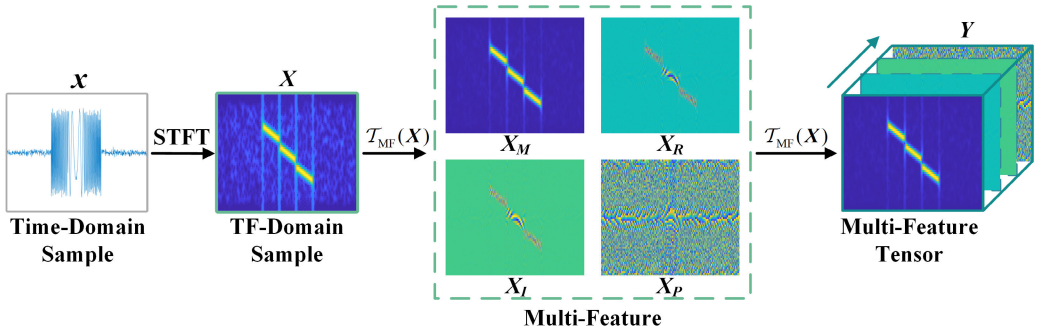


Fig. 3. Flowchart of SAR jamming data generation.

require balanced training samples as strictly as the probability-based methods.

Therefore, a jamming recognition strategy based on discriminative feature distance metric is proposed to solve imbalanced training samples. Specifically, we capture and refine modulation differences from various jamming TF spectrograms as discriminative features through a designed CNN $\mathcal{N}^{\theta}_{\mathrm{Net}}(\cdot;\theta)$. The intrinsic modulation differences increase the interclass distances of these features, i.e.,

$$\left\|\mathcal{N}^{\theta}_{\mathrm{Net}}(\boldsymbol{Y}_i;\theta)-\mathcal{N}^{\theta}_{\mathrm{Net}}\left(\boldsymbol{Y}_j;\theta\right)\right\|_2^2>\frac{1}{\eta},\quad \text{s.t. } \mathcal{C}\left(\boldsymbol{Y}_j\right)\neq\mathcal{C}(\boldsymbol{Y}_i) \tag{6}$$

where η is a number tends to 0, $\mathcal{N}^{\theta}_{\mathrm{Net}}(\boldsymbol{Y}_i;\theta),\mathcal{N}^{\theta}_{\mathrm{Net}}(\boldsymbol{Y}_j;\theta)\in\Gamma$ are discriminative features of jamming samples $\boldsymbol{Y}_i$ and $\boldsymbol{Y}_j$, $\mathcal{C}(\cdot)\in\mathbb{R}^{l\times 1}$ is the true label of $\boldsymbol{Y}\in\mathbb{R}^{t\times f\times 4}$, and $\|\cdot\|_2^2$ denotes the square of the vector L2-norm. Γ is embedding space consisting of discriminative features.

On the contrary, jamming samples of the same type have similar modulations, which reduce the intraclass distances of their discriminative features

$$\left\|\mathcal{N}^{\theta}_{\mathrm{Net}}(\boldsymbol{Y}_i;\theta)-\mathcal{N}^{\theta}_{\mathrm{Net}}(\boldsymbol{Y}_k;\theta)\right\|_2^2<\eta,\quad \text{s.t. } \mathcal{C}(\boldsymbol{Y}_k)=\mathcal{C}(\boldsymbol{Y}_i). \tag{7}$$

Thus, from (6) and (7), the proposed jamming recognition strategy via discriminative feature distance metric can be derived as

$$\begin{aligned}&\min_{\theta}\sum_{i=1}^{M}\left(\mathcal{D}^{+}-\mathcal{D}^{-}\right)\\ &\text{s.t.}\begin{cases}\mathcal{D}^{+}=\left\|\mathcal{N}^{\theta}_{\mathrm{Net}}(\boldsymbol{Y}_i;\theta)-\mathcal{N}^{\theta}_{\mathrm{Net}}(\boldsymbol{Y}_k;\theta)\right\|_2^2\\ \mathcal{D}^{-}=\left\|\mathcal{N}^{\theta}_{\mathrm{Net}}(\boldsymbol{Y}_i;\theta)-\mathcal{N}^{\theta}_{\mathrm{Net}}\left(\boldsymbol{Y}_j;\theta\right)\right\|_2^2\\ \mathcal{C}(\boldsymbol{Y}_k)=\mathcal{C}(\boldsymbol{Y}_i)\\ \mathcal{C}\left(\boldsymbol{Y}_j\right)\neq\mathcal{C}(\boldsymbol{Y}_i)\\ i=1,2,\ldots,M.\end{cases}\end{aligned} \tag{8}$$

Equation (8) optimizes the distribution of discriminative features formed by modulation differences, and allows CNN to directly learn the expression paradigm from jamming TF spectrograms to discriminative features. Notably, learning the expression paradigm in (8) is not limited by the balance of jamming samples. Thus, it naturally addresses the imbalanced training samples in real, noncooperative jamming scenarios. Meanwhile, various jamming signals are modulated from the same echo, which determines they are mapped to the same embedding space after (8). This essential property endows the proposed strategy to generally recognize multiple types of jamming within/outside the training set.

Under the constraint of (8), jamming type can be recognized by the nearest distance metric

$$\boldsymbol{C}=\arg\min_{c}\left\|\mathcal{N}^{\theta}_{\mathrm{Net}}\left(\tilde{\boldsymbol{Y}};\theta\right)-\mathcal{N}^{\theta}_{\mathrm{Net}}(\boldsymbol{Y}_c;\theta)\right\|_2^2,\ \text{s.t. } \boldsymbol{Y}_c\in\Phi_{\mathrm{qu}} \tag{9}$$

where $\boldsymbol{C}$ is the predicted jamming type, Φ_{qu} is the query dataset, consisting of a tiny quantity of jamming samples with known types.

B. Proposed Recognition Mechanism Extended for SAR Jamming Suppression

Capturing the modulation differences between jamming-jamming and jamming-target is not only the key to jamming recognition but also to jamming suppression [25], [26], [27]. Therefore, we innovatively apply the proposed jamming recognition strategy to SAR jamming suppression.

According to the mechanism in part A, once jamming is accurately recognized, the designed CNN has certainly captured the modulation differences of various jamming. These differences reflect the positions of the jamming signals in the TF spectrograms. Consequently, the jamming signal can be located by modulation differences captured in the shallow feature map $\boldsymbol{F}_{t\times f}$

$$\boldsymbol{M}_{t\times f}=\begin{cases}0, & \boldsymbol{R}_{\mathrm{jam}}\in\boldsymbol{F}_{t\times f}\\ 1, & \text{else}\end{cases} \tag{10}$$

where $\boldsymbol{R}_{\mathrm{jam}}$ and $\boldsymbol{M}_{t\times f}$ are the modulation difference region and corresponding jamming mask with a size of $t\times f$. Then, $\boldsymbol{M}_{t\times f}$ is used as a notch filter to suppress jamming components in TF spectrogram $\boldsymbol{X}(t,f)$

$$\hat{\boldsymbol{X}}(t,f)=\boldsymbol{X}(t,f)\otimes\boldsymbol{M}_{t\times f} \tag{11}$$

where $\hat{\boldsymbol{X}}(t,f)$ is the jamming-suppressed TF spectrogram and $\otimes$ is the dot product. Thus far, SAR jamming suppression has been achieved synchronously only by extending the proposed jamming recognition mechanism, which undoubtedly reduces resource consumption.

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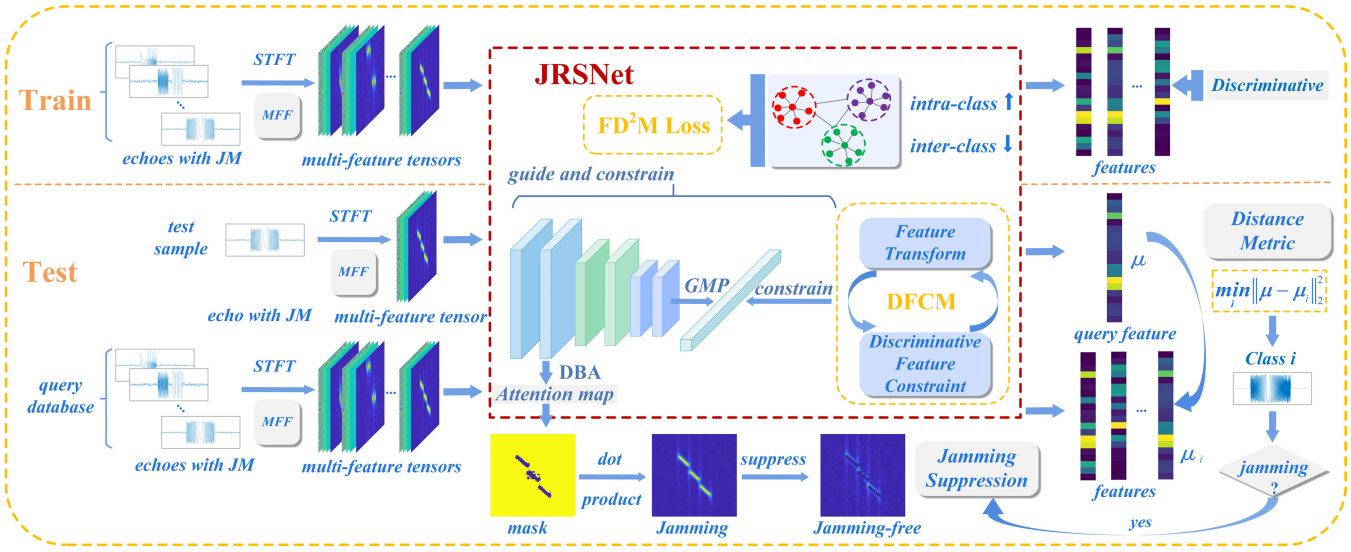


Fig. 4. Overall processing of the proposed jamming recognition strategy. The designed JRSNet refines the jamming modulation differences in TF spectrograms and forms discriminative features. FD^{2}M loss and DFCM guarantee these processes. The jamming type can be recognized by the nearest feature distance metric. And jamming suppression can be realized synchronously.

IV. Proposed Jamming Recognition Network

This section specifically implements the jamming recognition mechanism in Section III. The following first starts with an overview. Then, the whole structure and several key components of the derived JRSNet are elaborated.

A. Overview

The processing flow of the proposed jamming recognition strategy under imbalanced training samples is depicted in Fig. 4. First, the STFT and MFF are performed sequentially to jamming echoes to obtain MFTs. Then, the designed JRSNet captures jamming modulation differences from these tensors and refines them into discriminative features. These processes are guided and constrained by FD^{2}M loss function and DFCM and form a feature expression paradigm in (8). After that, the final jamming type is predicted by the nearest feature distance metric in (9). Meanwhile, the jamming signal is suppressed synchronously by (11).

B. JRSNet

Following the proposed jamming recognition strategy in Section III, it is anticipated that the network can fully extract jamming modulation differences in TF spectrograms, and then refine discriminative features of various jamming. Meanwhile, the whole jamming recognition process should be without balanced training samples. Thus, JRSNet is derived to achieve these goals.

As shown in Fig. 5, JRSNet mainly consists of a CAA block and a DFCM, two DBA blocks, and three convolution-pooling blocks. Attention-based DBA and CAA are designed for capturing and refining jamming modulation information in TF spectrograms from multiple dimensions. Therefore, they are embedded after the convolution-pooling block. The refined modulation information undergoes global max pooling (GMP) to output the final jamming feature $\boldsymbol{\mu} \in \mathbb{R}^{n\times 1}$ for recognition.

TABLE I
Specific Parameters of JRSNet

Layer	Conv. Kernel	Pooling	Activation	Normalization
Input		128×960×4		
Conv.1	(3×3)×8	2×2	Leaky ReLU	BN
DBA.1	(1×1)×16	-	Leaky ReLU	BN
	(3×3)×8	-	Leaky ReLU	BN
Conv.2	(3×3)×16	2×2	Leaky ReLU	BN
DBA.2	(1×1)×16	-	Leaky ReLU	BN
	(3×3)×8	-	Leaky ReLU	BN
Conv.3	(3×3)×20	2×2	Leaky ReLU	BN
GMP	-	Global	-	-
GAP	-	Global	-	-
CAA	(1×1)×20	-	Sigmoid	-
DFCM		1×1×c		
Output		1×20		

Meanwhile, DFCM ensures the discriminability of $\boldsymbol{\mu}$ through feature transformation.

The specific structure and parameters of JRSNet are listed in Table I. In addition, JRSNet is trained under the guidance of the proposed FD^{2}M loss function, which is the prerequisite for solving imbalanced training samples. Thus, the obtained results can be concluded as

$$\mathcal{N}_{\mathrm{JRS}}^{\theta} = \underset{\mathrm{FD^2M}}{\arg\min}\{\boldsymbol{\mu}|\mathcal{N}_{\mathrm{JRS}}, \theta\}. \tag{12}$$

C. DBA and CAA

Jamming modulation information manifests itself as distinctly anomalous textures in TF spectrograms. Capturing them requires JRSNet to focus on salient regions in TF spectrograms. Therefore, DBA and CAA are designed to solve this issue, and they are attention-based.

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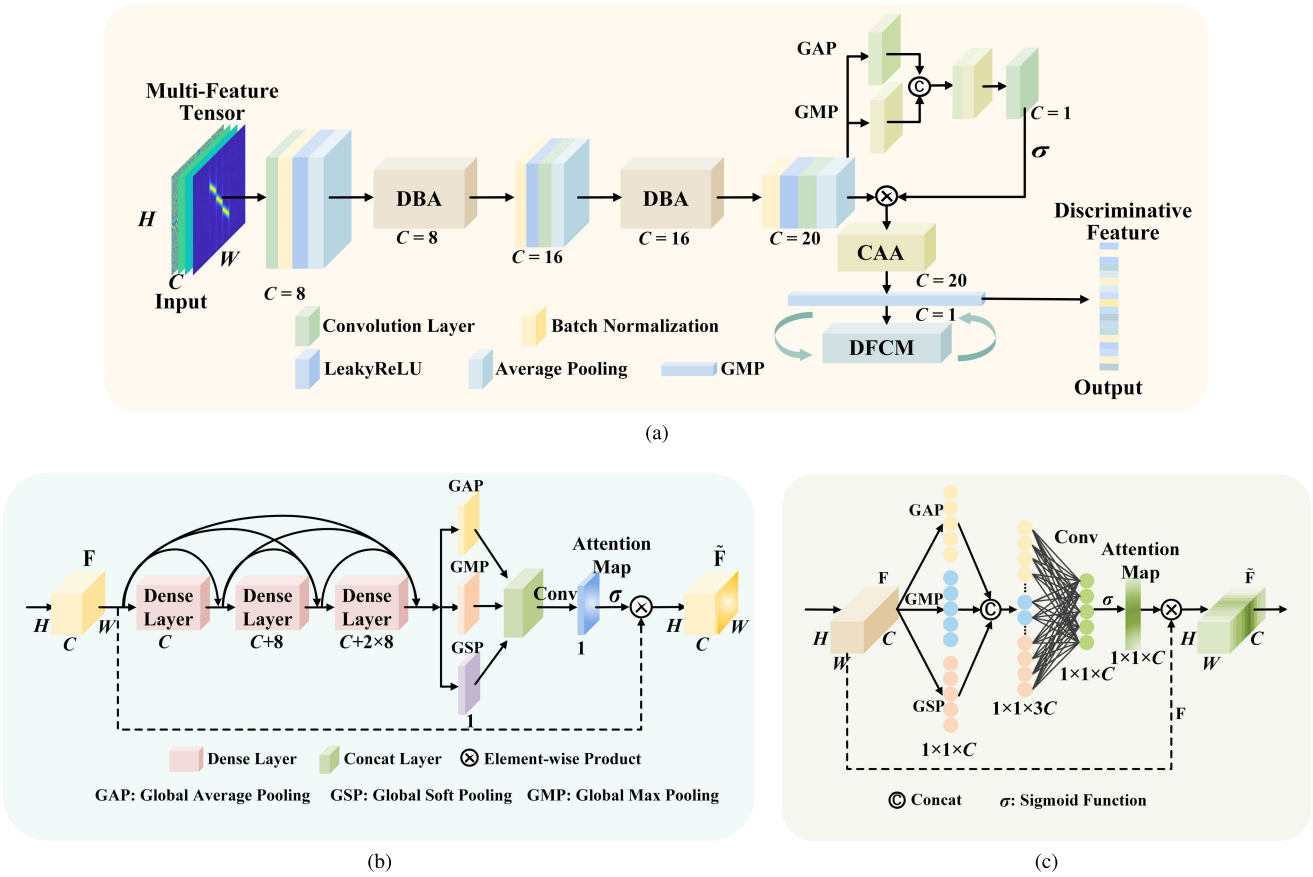


Fig. 5. Structures of the proposed (a) JRSNet, (b) DBA module, and (c) CAA module.

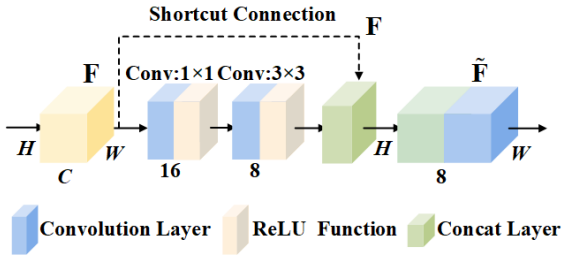


Fig. 6. Structure of the dense layer.

Figs. 5 and 6 show the structure of the DBA. The dense layer is utilized to reuse multilayer features, facilitating the flow of complex jamming modulation information in JRSNet. Global soft pooling (GSP) $g_s(\cdot)$ [32] is specifically introduced on top of GMP $g_m(\cdot)$ and global average pooling (GAP) $g_a(\cdot)$. The purpose is to capture the jamming modulation information in feature maps, and then form a more refined jamming attention map $\boldsymbol{A}$. The proposed spatial DBA can be formulated as

$$\hat{\boldsymbol{Z}} = \sigma\left\{\boldsymbol{W}_{7\times7}^{\mathrm{T}}\left[g_a(\boldsymbol{D}); g_m(\boldsymbol{D}); g_s(\boldsymbol{D})\right]\right\} \otimes \boldsymbol{Z} = \boldsymbol{A} \otimes \boldsymbol{Z} \tag{13}$$

where $\boldsymbol{D}$ is the output of dense layer, $\boldsymbol{W}$ denotes the convolution kernel weight with size of 7×7, $\sigma(\cdot)$ is the sigmoid function, and $\boldsymbol{Z}$ and $\hat{\boldsymbol{Z}}$ are the input and the refined feature, respectively.

Since multiple convolutions operate, the jamming modulation information is also diffused in several channels. Thus, the CAA is similarly designed to gather this information from the channel dimension

$$\hat{\boldsymbol{F}} = \sigma\left\{\boldsymbol{W}_{1\times1}^{\mathrm{T}}\left[g_a(\boldsymbol{F}); g_m(\boldsymbol{F}); g_s(\boldsymbol{F})\right]\right\} \otimes \boldsymbol{F} \tag{14}$$

where $\boldsymbol{W}$ is the convolution kernel weight with size of 1×1, and $\boldsymbol{F}$ and $\hat{\boldsymbol{F}}$ denote the feature maps before and after CAA.

DBA and CAA capture jamming modulation information from multiple dimensions and form it into a refined discriminative feature. Thereby, this purified feature can be used to improve recognition performance. Moreover, the jamming attention map generated by DBA can be segmented to produce the jamming mask in (10), thus synchronously suppressing the jamming signal.

D. Discriminative Feature Constraint Module

Final jamming features need to be ensured that they are discriminative before recognition. Most directly related to discriminability is the jamming class label, which generally forms one-hot probabilistic coding $\boldsymbol{C} = [0, 0, \ldots, 1, \ldots, 0]^{\mathrm{T}} \in \mathbb{R}^{c\times1}$ (c is the number of classes). To establish connections with jamming class label, the final feature $\boldsymbol{\mu} \in \mathbb{R}^{n\times1}$ is first transformed into the class space $\mathbb{R}^{c\times1}$

$$\boldsymbol{v} = \boldsymbol{W}\boldsymbol{\mu} + \boldsymbol{b} \tag{15}$$

where $\boldsymbol{W} \in \mathbb{R}^{c\times n}$ and $\boldsymbol{b} \in \mathbb{R}^{c\times1}$ are the transformation matrix and bias, respectively, and $\boldsymbol{v} \in \mathbb{R}^{c\times1}$ is the transformed feature.

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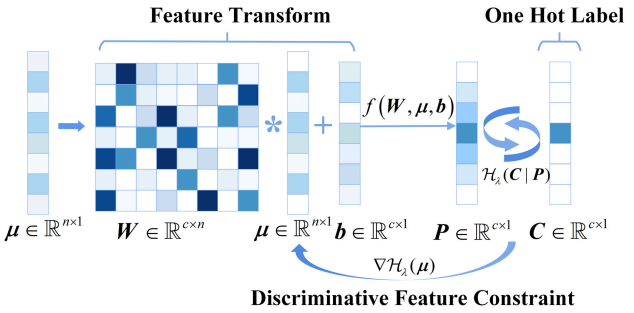


Fig. 7. Structure of the proposed DFCM.

Inspired by the T-softmax, $\boldsymbol{v}$ is mapped to the probability interval

$$p_i = f(\boldsymbol{v}) = f(\boldsymbol{W},\boldsymbol{\mu},\boldsymbol{b}) = \frac{\exp(v_i/\lambda)}{\sum_{j=1}^{n}\exp(v_j/\lambda)} = \frac{\exp\{(\boldsymbol{w}_i\boldsymbol{\mu}+\boldsymbol{b}_i)/\lambda\}}{\sum_{j=1}^{n}\exp\{(\boldsymbol{w}_j\boldsymbol{\mu}+\boldsymbol{b}_j)/\lambda\}} \tag{16}$$

$$\boldsymbol{P} = \begin{bmatrix} p_1 & p_2 & \cdots & p_c \end{bmatrix}^{\mathrm{T}} \tag{17}$$

where $v_i (i = 1, 2, \ldots, c)$ is each term in $\boldsymbol{v}$, $\lambda = 1/c$ denotes temperature parameter, p_i is the probability of v_i, $\boldsymbol{w}_i$ is a row vector in transformation matrix $\boldsymbol{W}$, and $\boldsymbol{P}$ is the predicted class probability vector.

On this basis, T-cross entropy $\mathcal{H}_\lambda(\boldsymbol{C}|\boldsymbol{P})$ is utilized to measure the extent to which $\boldsymbol{P}$ approximates $\boldsymbol{C}$

$$\mathcal{H}_\lambda(\boldsymbol{C}|\boldsymbol{P}) = -\sum_{i=1}^{c} c_i \ln p_i \tag{18}$$

the smaller $\mathcal{H}_\lambda(\boldsymbol{C}|\boldsymbol{P})$ is, the closer $\boldsymbol{P}$ to $\boldsymbol{C}$, and the more related the jamming feature $\boldsymbol{\mu}$ to the jamming class label, the stronger its discriminability. The key to finding the optimal solution $\boldsymbol{\mu}^*$ of minimum $\mathcal{H}_\lambda(\boldsymbol{C}|\boldsymbol{P})$ lies in updating $\boldsymbol{W}$, which conforms to backpropagation with $\mathcal{H}_\lambda(\boldsymbol{C}|\boldsymbol{P})$ as the loss function. Consequently, $\mathcal{H}_\lambda(\boldsymbol{C}|\boldsymbol{P})$ is added as an item to the FD^{2}M loss function in part E to enhance the discriminability of $\boldsymbol{\mu}$. Fig. 7 illustrates the structure of the proposed DFCM. After the DFCM, the jamming feature $\boldsymbol{\mu}$ becomes discriminative, which facilitates accurate jamming recognition.

E. FD^{2}M Loss Function

Although the above modules enable JRSNet to extract jamming discriminative features, the influence of imbalanced training samples remains. As analyzed in Section III, to eliminate this influence, JRSNet needs to learn the expression paradigm from TF spectrograms to discriminative features. Therefore, the FD^{2}M loss function is defined to constrain JRSNet.

Based on (8), we can easily derive the relative distance loss function $\mathcal{L}_{\mathrm{rd}}$

$$\mathcal{L}_{\mathrm{rd}} = \frac{1}{M}\sum_{i=1}^{M}\left\{d\left(\boldsymbol{\mu}_{\boldsymbol{Y}_i},\boldsymbol{\mu}_{\boldsymbol{Y}_i^p}\right) - d\left(\boldsymbol{\mu}_{\boldsymbol{Y}_i},\boldsymbol{\mu}_{\boldsymbol{Y}_i^n}\right)\right\}_+ \tag{19}$$

where

$$d\left(\boldsymbol{\mu}_{\boldsymbol{Y}_i},\boldsymbol{\mu}_{\boldsymbol{Y}_i^p}\right) = \left\|\boldsymbol{\mu}_{\boldsymbol{Y}_i} - \boldsymbol{\mu}_{\boldsymbol{Y}_i^p}\right\|_2^2 \tag{20}$$

$$d\left(\boldsymbol{\mu}_{\boldsymbol{Y}_i},\boldsymbol{\mu}_{\boldsymbol{Y}_i^n}\right) = \left\|\boldsymbol{\mu}_{\boldsymbol{Y}_i} - \boldsymbol{\mu}_{\boldsymbol{Y}_i^n}\right\|_2^2 \tag{21}$$

these two terms are the intraclass distance and interclass distance of discriminative features, respectively. $\{\cdot\}_+$ is the part greater than 0. $\boldsymbol{\mu}_{\boldsymbol{Y}_i^p}$ and $\boldsymbol{\mu}_{\boldsymbol{Y}_i^n}$ denote jamming discriminative features of the same and different classes as $\boldsymbol{\mu}_{\boldsymbol{Y}_i}$, respectively.

As (19) optimizes the relative distance of each jamming sample, the computation increases significantly. Hence, we substitute merely the boundary condition of (19) for itself. After that, the relative distance loss function is

$$\mathcal{L}_{\mathrm{rd}} = \frac{1}{M}\sum_{i=1}^{M}\left\{\frac{1}{\lambda}\max_p d\left(\boldsymbol{\mu}_{\boldsymbol{Y}_i},\boldsymbol{\mu}_{\boldsymbol{Y}_i^p}\right) - \lambda\min_n d\left(\boldsymbol{\mu}_{\boldsymbol{Y}_i},\boldsymbol{\mu}_{\boldsymbol{Y}_i^n}\right)\right\}_+. \tag{22}$$

Furthermore, the compactness of intraclass jamming features needs to be restricted. It prevents the distribution of these features from being too discrete to be recognized. Therefore, we use R as the radius of class center $\boldsymbol{c}_{\mathcal{C}(\boldsymbol{Y}_i)}$, and design the center distance loss $\mathcal{L}_{\mathrm{cd}}$ to confine the jamming feature $\boldsymbol{\mu}_{\boldsymbol{Y}_i}$ to the hypersphere range

$$\mathcal{L}_{\mathrm{cd}} = R^2 + \left\{\frac{1}{M}\sum_{i=1}^{M}\left\|\boldsymbol{\mu}_{\boldsymbol{Y}_i} - \boldsymbol{c}_{\mathcal{C}(\boldsymbol{Y}_i)}\right\|_2^2 - R^2\right\}_+ \tag{23}$$

where R^2 and $\boldsymbol{c}_{\mathcal{C}(\boldsymbol{Y}_i)}$ descend automatically with the training process until they converge. Jamming features will become more clustered by (23), which facilitates identifying jamming types by the nearest distance metric.

Combining the analysis of feature discriminability constraints in Part D, the FD^{2}M loss function proposed in this article is

$$\mathcal{L}_{\mathrm{FD^2M}} = \mathcal{L}_{\mathrm{rd}} + \beta\mathcal{L}_{\mathrm{cd}} + \mathcal{H}_\lambda(\boldsymbol{C}|\boldsymbol{P}) \tag{24}$$

where $\beta = 0.01$ is used to control the proportion of $\mathcal{L}_{\mathrm{cd}}$.

Equation (24) realizes and further optimizes the expression paradigm in (8) unrelated to training sample balance. Guided by (24), JRSNet learns directly the expression from jamming TF spectrograms to discriminative features. After that, the jamming type is measured by the extracted discriminative features. This significantly improves the recognition accuracy under imbalanced training samples. Moreover, as training progresses, JRSNet maps various jamming discriminative features into the same embedding space, enhancing its own generalization.

The framework of JRSNet for SAR jamming recognition and suppression is shown in Algorithm (1).

V. Experimental Results

In this section, the SAR jamming datasets and implementation of JRSNet are first detailed. Next, the extracted jamming features are visualized to illustrate their discriminability. Then, the recognition performance of JRSNet under imbalanced training samples is verified on various SAR jamming data. After that, unique jamming suppression experiments are conducted via JRSNet. Finally, ablation results are presented, and the necessity of each component in JRSNet is analyzed.

A. Experimental Data

To verify the performance of JRSNet, one target signal and nine jamming signals (1024 time points per signal) are utilized.

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Algorithm 1 JRSNet for SAR Jamming Recognition and Suppression Under Imbalanced Training Samples

Train JRSNet:

1: Initialize: jamming pulse X_i, learning rate = 1×10^{-3}, Max epoch = 40
2: Input: $Y_i = STFT(X_i)$, $\boldsymbol{Y} = \{Y_1, Y_2, \ldots, Y_N\}$, $N = 3000$
3: Process $\boldsymbol{Y}$ via (3), and the training proportion is set to Table III
4: **while** $j <$ Max epoch **do**
5: $j \leftarrow j + 1$
6: Get discriminative feature $\boldsymbol{\mu}_{\boldsymbol{Y}_i}$ using JRSNet
7: Optimize loss $\mathcal{L}_{\text{FD}^2\text{M}}$ via (19)–(24) using the Adam
8: Backpropagation $\nabla\mathcal{L}_{\text{FD}^2\text{M}}$ and update parameters θ
9: **end while**
10: Obtain the trained JRSNet(θ^*) with the optimal parameters θ^*

Test JRSNet:

1: Initialization and input are the same as training JRSNet
2: Calculate jamming type c via (9) and the trained JRSNet(θ^*)
3: Output attention map from first DBA and suppress jamming by (11)

TABLE II
PARAMETERS OF JAMMING SIGNALS

ID	Signal	Parameters	Range of values
C1	Target	Bandwidth	300 MHz
		Sampling Frequency	360 MHz
C2	ISFJ	Number of forwards	1
		Samples	4
		Jamming Duty Cycle	0.5
C3	ISRJ	Number of forwards	2~3
		Samples	2
		Jamming Duty Cycle	0.5~0.75
C4	CSJ	Number of combs	8
C5	SSFJ	Number of subsections	3
		Frequency Modulation Slope	-1.7×10^{14}
C6	SRSFJ	Number of subsections	5
C7	NAMJ	Standard Deviation	1
		Bandwidth	200 MHz
C8	SRSFJ+ISRJ	Same as specific jamming	Same as specific jamming
C9	CSJ+ISFJ		
C10	NAMJ+SSFJ		

The jamming signals contain six single types: interrupted sampling and forwarding jamming (ISFJ) [14], ISRJ [31], comb spectrum jamming (CSJ), SSFJ [29], subsection random shift frequency jamming (SRSFJ) [29], noise amplitude modulation jamming (NAMJ) (simulated + measured) [30], and three composite types: SRSFJ + ISRJ, CSJ + ISFJ, and NAMJ + SSFJ, whose specific parameters are listed in Table II. Each jamming signal varies jamming-to-signal-ratio (JSR) from 5 to 30 dB at SNR = 0 dB. All signals are converted to the TF domain by STFT, and its corresponding parameters are overlap = 64, window type = hamming, window length = 65, and nfft = 128. Fig. 8 shows the TF spectrograms of each signal. These TF spectrograms eventually follow (3) to together form the datasets, and then they are normalized by min-max scaling to maintain the distribution

$$\tilde{\boldsymbol{X}} = \frac{\boldsymbol{X} - \boldsymbol{X}_{\min}}{\boldsymbol{X}_{\max} - \boldsymbol{X}_{\min}} \tag{25}$$

where $\boldsymbol{X}_{\max}$ and $\boldsymbol{X}_{\min}$ are the maximum and minimum values of $\boldsymbol{X}$, and $\tilde{\boldsymbol{X}}$ is the normalized MFT input to JRSNet.

TABLE III
TRAINING DATASET WITH IMBALANCED JAMMING SAMPLES

Signal	Number	Proportion
Target	500	16.67%
ISRJ	1000	33.33%
SRSFJ	200	6.67%
CSJ+ISFJ	300	10.00%
NAMJ+SSFJ	1000	33.33%

TABLE IV
TRAINING PARAMETERS OF JRSNET

Parameters	Values
Optimizer	Adam
Learning Rate	1×10^{-3}
Epochs	40
Batch Size	64
Input Size	128×960×4
Initial Radius	2.0

TABLE V
SPECIFIC PARAMETERS OF 1D-CNN

Layer	Conv. Kernel	Pooling	Activation	Dropout
Input	1×1024			
Conv.1	(5×1)×16	2×1	ReLU	-
Conv.2	(7×1)×32	2×1	ReLU	0.5
Conv.3	(9×1)×64	2×1	ReLU	0.5
GAP	1×64			
Linear	-	-	Softmax	-
Output	1×5			

B. Implementation Details

1) Experimental Settings: To construct an imbalanced training environment for JRSNet, the training dataset consists of five types of jamming samples in the proportions in Table III, all of them with a JSR of 30 dB. The whole training is performed on a GTX3090 graphics card and PyTorch framework, and the specific training parameters are listed in Table IV. Every 20 epochs learning rate decays 0.1, and R in (23) sets an initial value of 2.0. The training loss and radius curves are presented in Fig. 9. In testing, we examine all jamming signals (within and outside the training dataset) in Table II with a JSR of 5~25 dB, and the sample size of each type is 400. In addition, we average ten independent experiment results as the final result to ensure reliability.

2) Comparison Methods: We compare JRSNet with other jamming recognition methods including SVM, 1D-CNN, VGG16-A [29], WECNN [14], and JR-TFSAD [31]. The SVM algorithm utilizes the radial basis function (RBF) as a

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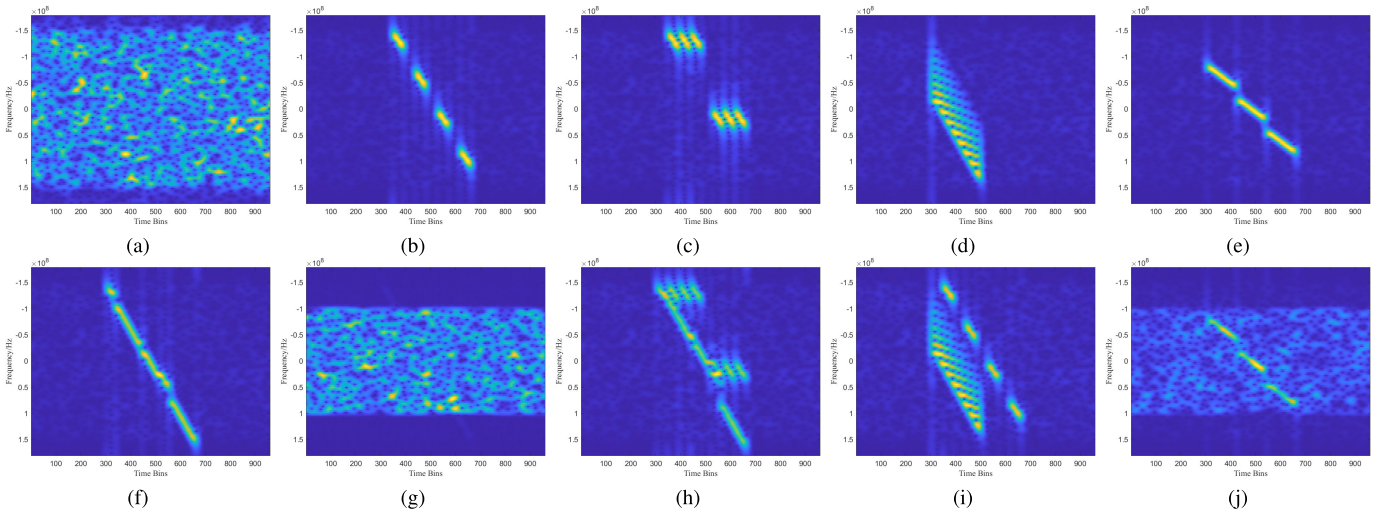


Fig. 8. TF spectrograms of ten signals. (a) Scene target. (b) ISFJ. (c) ISRJ. (d) CSJ. (e) SSFJ. (f) SRSFJ. (g) NAMJ. (h) SRSFJ + ISRJ. (i) CSJ + ISFJ. (j) NAMJ + SSFJ.

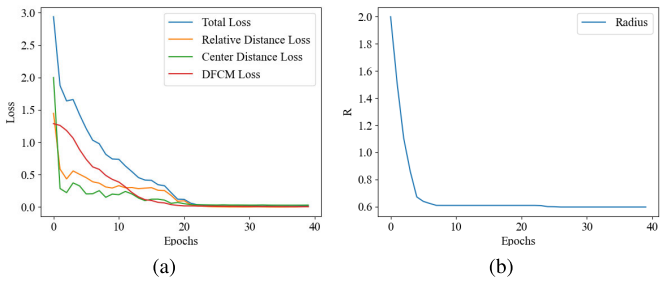


Fig. 9. Curves of (a) training loss and (b) radius, which converge as the training number increases.

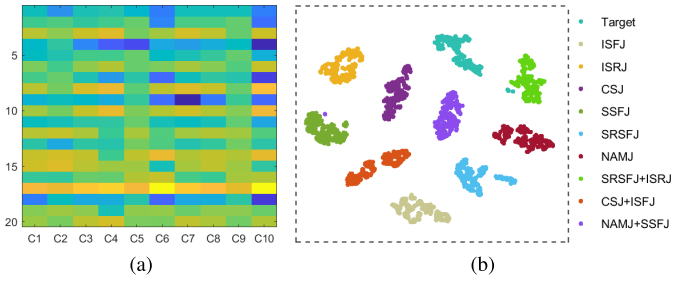


Fig. 10. Discriminative features visualization at JSR = 20 dB for ten signals. (a) Discriminative features of ten signals. (b) Features distribution in embedding space.

kernel function, and its input features include mean, variance, moment kurtosis, moment skewness, information entropy, and box dimension. 1D-CNN is constructed on the time-domain jamming data with parameters in Table V. The structure of other methods is consistent with the corresponding literature. Among them, both WECNN and JR-TFSAD use TF spectrograms as input, while VGG16-A employs SAR images.

C. Evaluation Metrics

The performance of JRSNet is evaluated in terms of jamming recognition, jamming suppression, and running time. Specifically, recall, accuracy, precision, $F1$ score, and Kappa are introduced to quantitatively measure the recognition performance. These indicators are determined by the confusion matrix [33], which all range from 0 to 1. The higher these indicators, the more accurate the recognition. To validate the unique jamming suppression performance of JRSNet, the signal distortion ratio (SDR) [4] and the structure similarity (SSIM) [12] are chosen. SDR is a ratio of the power loss after jamming suppression to the signal power. SSIM measures the similarity between jamming-free SAR images and jamming-suppressed SAR images with an image size of 430 × 1024. A smaller SDR and a higher SSIM (with a range of 0∼1) indicate a better suppression performance. In addition, the testing time is measured to reflect the real-time performance of JRSNet.

D. Visualization of the Discriminative Feature

To illustrate the discriminability of the features extracted by JRSNet, we intuitively visualize the features of the signals in Table II at JSR = 20 dB, as shown in Fig. 10. Each column in Fig. 10(a) represents a type of jamming feature. There are notable differences across different jamming features. It is primarily attributed to DBA and CAA, which adaptively capture jamming modulation differences in TF spectrograms, and then form refined features. We apply t-SNE [34] to observe the distribution of these refined features in the embedding space Γ, as shown in Fig. 10(b). In Γ, these refined features exhibit less intraclass variation and greater interclass separation. This indicates that constrained by DFCM and FD^{2}M loss, these features have strong discriminability, which is critical for jamming recognition under imbalanced training samples. Encouragingly, Fig. 10 implies a high generalization of JRSNet, as it separates jamming types uninvolved in training in the same embedding space.

E. Results of Jamming Recognition Under Imbalanced Training Samples

In this section, JRSNet recognizes jamming types based on (9), where the query and test datasets consist of 10 and 400 samples for each signal in Table II, respectively.

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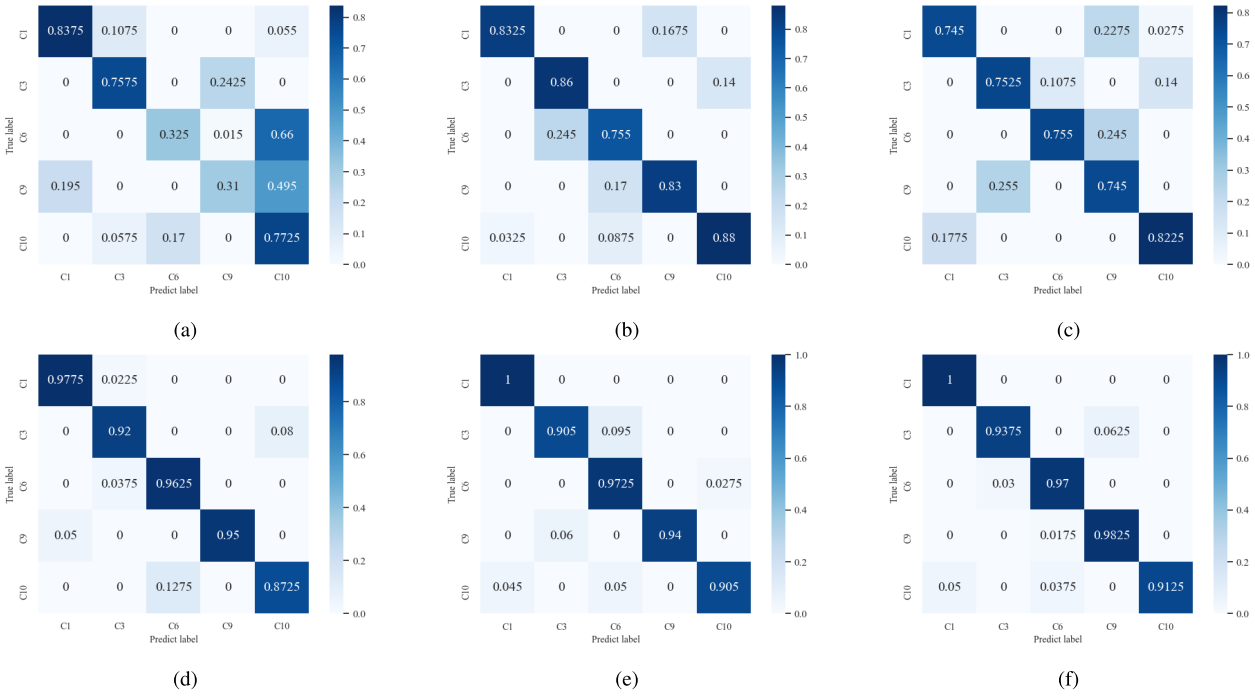


Fig. 11. Confusion matrix at JSR = 20 dB between JRSNet and other comparison methods under imbalanced training samples. (a) SVM. (b) 1D-CNN. (c) VGG16-A. (d) WECNN. (e) JR-TFSAD. (f) JRSNet (ours).

TABLE VI

RECOGNITION RESULTS OF JAMMING AT JSR = 20 dB FOR JAMMING TYPE WITHIN THE TRAINING DATASET UNDER IMBALANCED TRAINING SAMPLES

Precision(%)↑	SVM	1D-CNN	VGG16-A	WECNN	JR-TFSAD	JRSNet(ours)
Target	81.11±4.82	86.13±2.02	80.67±3.15	95.13±0.21	**95.65±0.42**	95.12±0.21
ISRJ	76.32±3.91	77.31±2.43	74.62±4.22	93.88±0.60	93.78±0.67	**96.82±0.14**
SRSFJ	36.97±5.93	73.93±2.55	87.54±3.89	88.30±1.01	89.02±1.24	**94.20±0.78**
CSJ+ISFJ	31.08±3.66	82.99±1.95	61.19±4.62	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
NAMJ+SSFJ	70.69±4.85	86.75±2.63	83.08±3.75	91.60±0.89	97.05±0.94	**100.00±0.00**
Accuracy(%)↑	52.18±4.02	82.72±2.36	76.40±4.10	93.65±0.44	94.45±0.80	**95.95±0.28**
Recall(%)↑	60.18±5.28	82.85±3.11	76.52±3.92	93.78±0.39	94.71±0.75	**95.93±0.30**
F1 score(%)↑	59.21±4.15	82.83±2.85	76.68±3.55	93.60±0.62	94.47±0.54	**94.88±0.27**
Kappa(%)↑	50.06±3.54	78.39±3.47	70.50±3.49	92.06±0.48	93.06±0.68	**95.93±0.22**

1) Recognition for Jamming Type Within the Training Dataset: This section first compares the recognition performance of five jamming signals in Table III. Fig. 11 shows the confusion matrix of multiple methods trained on imbalanced samples with JSR = 20 dB, where C1∼C10 correspond to Table II. Compared to the CNN-based methods, SVM confuses a large number of SRSFJ and CSJ + ISFJ with other signals and obtains a low average accuracy of 60%. The time-domain signals input for 1D-CNN have fewer modulation differences than TF spectrograms. Hence, 1D-CNN results in less recognition accuracy than WECNN, JR-TFSAD, and JRSNet about 10%. ISRJ, SRSFJ, and CSJ + ISFJ all appear as multiple false targets in SAR images, making it hard for VGG16-A to distinguish them based on the image domain. As a result, its average accuracy is only about 76%. At high JSR, WECNN, JR-TFSAD, and JRSNet all achieve more than 90% recognition accuracy.

Table VI provides comprehensive recognition results of the different methods. SVM obtains precision lower than CNN-based methods 28%∼45% owing to the unstable classification surface learned from imbalanced training samples. WECNN, JR-TFSAD, and JRSNet all accurately identify these five types of jamming samples; in particular, the recognition precision of CSJ + ISFJ reaches 100%. However, influenced by the imbalanced training data, WECNN and JR-TFSAD vary recognition precisions from 88.30% to 100%, which means poor robustness of these methods. Despite the random frequency shift of SRSFJ making recognition more difficult, JRSNet still reaches 94.20% recognition precision with lower variance. This is attributed to that JRSNet powerfully captures jamming modulation information to form discriminative features, and then predicts jamming type by a distance metric. Notably, JRSNet outperforms JR-TFSAD, in terms of accuracy, recall, $F1$ score, and kappa by 1.5%, 1.2%, 0.41%, and 2.87%, respectively. It is indicated that JRSNet can accurately recognize multiple jamming types with strong robustness under imbalanced training samples.

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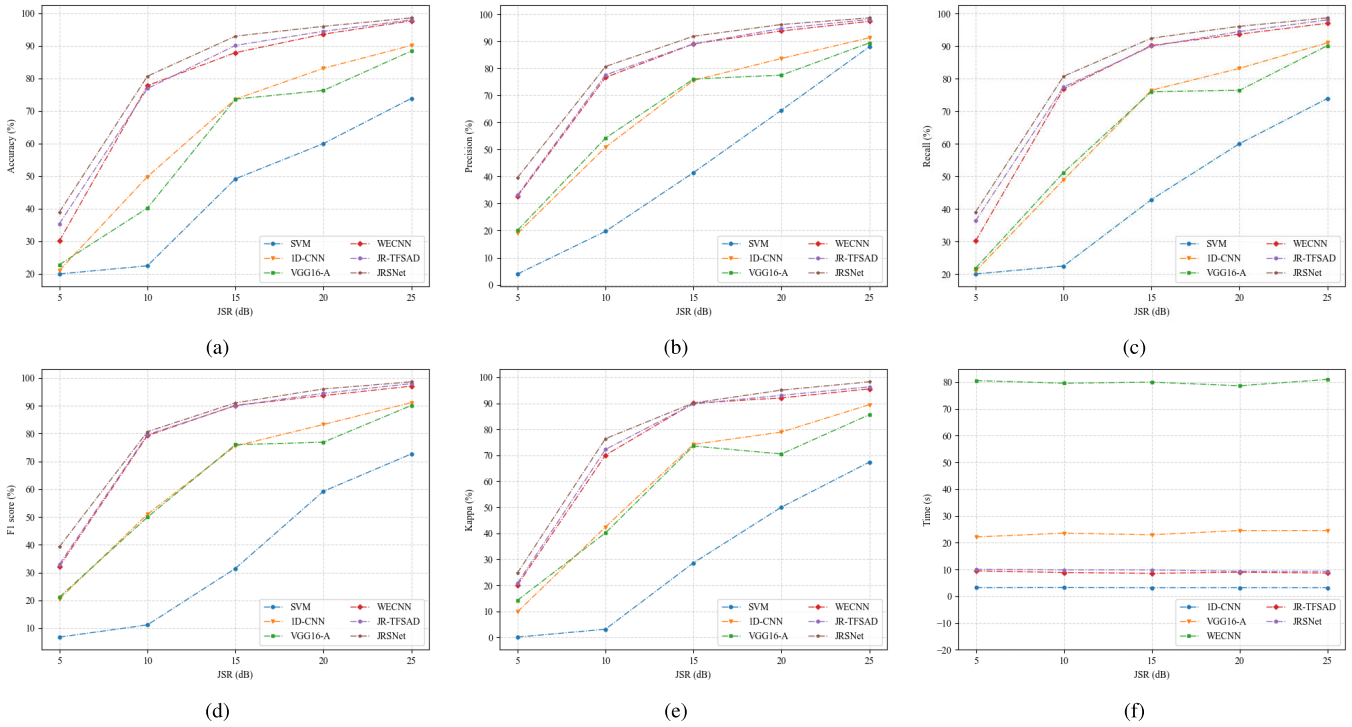


Fig. 12. Comparison of the comprehensive metrics of different methods for jamming recognition. (a) Accuracy. (b) Precision. (c) Recall. (d) $F1$ score. (e) Kappa. (f) Time.

TABLE VII

COMPARISON OF THE AVERAGE TESTING TIME OF DIFFERENT METHODS FOR JAMMING RECOGNITION

Methods	Running Time(s)↓
SVM	-
1D-CNN	**3.86±0.29**
VGG16-A	22.63±1.27
WECNN	80.31±1.65
JR-TFSAD	9.52±0.54
JRSNet(ours)	9.87±0.16

TABLE VIII

AVERAGE RECOGNITION RESULTS FOR JAMMING TYPE OUTSIDE THE TRAINING DATASET UNDER IMBALANCED TRAINING SAMPLES

Jamming	Precision(%)↑	Recall(%)↑	F1 score(%)↑	Time(s)↓
ISFJ	**94.86±1.02**	94.22±0.43	94.74±0.35	9.63±0.35
CSJ	94.32±0.62	93.53±0.24	**94.95±0.31**	**9.35±0.24**
SSFJ	93.16±0.46	**94.46±0.27**	93.76±0.19	9.68±0.30
NAMJ	87.89±0.55	93.15±0.11	90.41±0.21	9.43±0.38
SRSFJ+ISRJ	94.26±0.62	89.03±0.28	91.27±0.43	9.55±0.47
Accuracy(%)		92.82±0.31		
Kappa(%)		91.03±0.25		

TABLE IX

RECOGNITION RESULTS OF MEASURED NAMJ

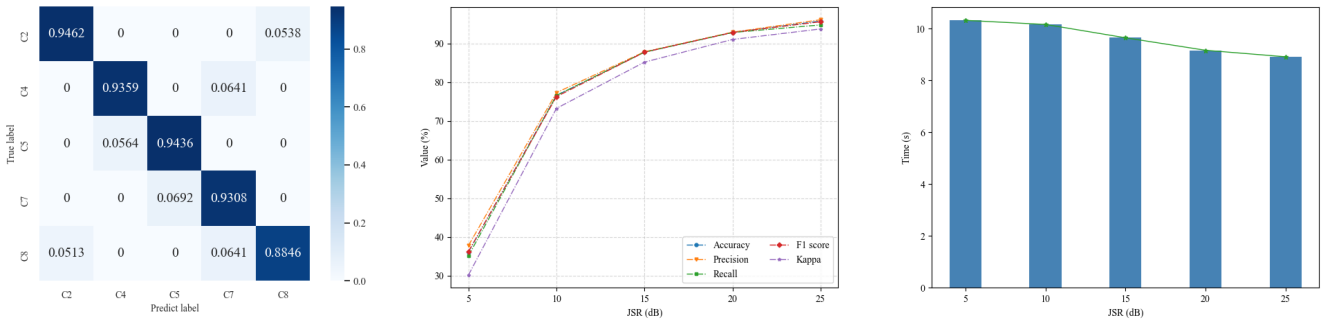
Jamming	Precision(%)↑	Recall(%)↑	F1 score(%)↑	Accuracy (%)↑
NAMJ	94.53±0.50	93.08±0.21	93.81±0.39	93.85±0.42

To compare generalization, we identify jamming samples uninvolved in training, and the JSRs are 5, 10, 15, 20, and 25 dB, as shown in Fig. 12. When JSR $\geq$ 15 dB, the accuracy, precision, recall, $F1$ score, and kappa of all the methods are raised as JSR increases, and the average recognition accuracy can reach about 80%. However, once JSR reduces to 10 dB or 5 dB, the jamming signals overlap with the background in TF spectrograms, and while the recognition performance of other methods decreases significantly, JRSNet still outperforms other methods by 5%∼20% in recognition indicators. This means that JRSNet can maintain generalization across multiple jamming recognition scenarios.

Fig. 12(f) and Table VII compare the testing time of CNN-based methods. It can be seen that at various JSRs, each method consumes nearly the same testing time. Since the input size of 1D-CNN is 1024 × 1, much smaller than 128 × 960, the testing time of 1D-CNN is the shortest of 3.86 s. WECNN integrates four CNN models with high complexity and consumes the longest testing time of 80.31 s. The testing time of JRSNet and JR-TFSAD is almost the same (more than 9 s), but JRSNet performs better in jamming comprehensive recognition.

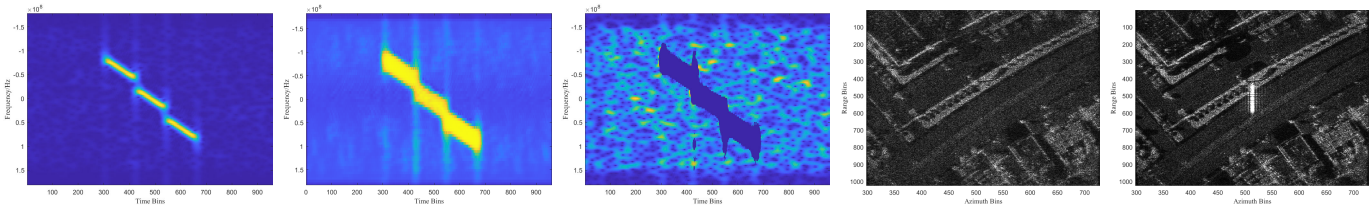
2) Recognition for Jamming Type Outside the Training Dataset: SVM, 1D-CNN, VGG16-A, WECNN, and JR-TFSAD are probability-based and unable to identify jamming types not involved in the training. Consequently, this section only evaluates the performance of JRSNet, and the results are presented in Fig. 13 and Table VIII. From Table VIII, even though these five types of jamming are not involved in training, JRSNet still reaches an average accuracy and kappa of 92.82% and 91.03%, respectively. Moreover, the recognition accuracy of JRSNet in Table IX for measured NAMJ also exceeds 90%. It demonstrates that JRSNet fully learns the expression paradigm from jamming

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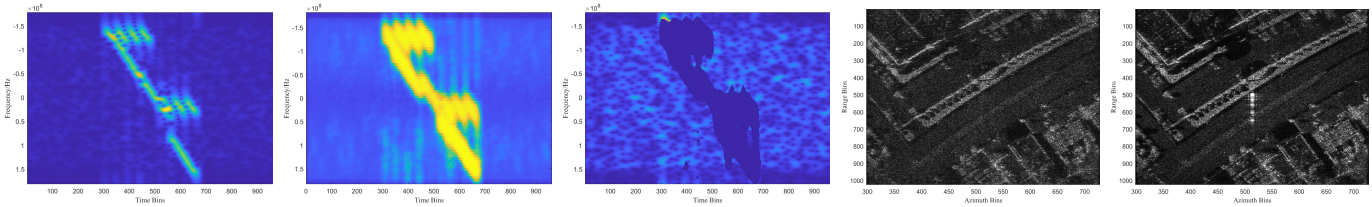
(a) (b) (c)

Fig. 13. Average recognition performance for jamming type outside training dataset at JSR = 20 dB. (a) Confusion matrix. (b) Recognition metrics. (c) Testing time.



(a) (b) (c) (d) (e)

Fig. 14. Jamming suppression results of SSFJ. (a) SSFJ-contaminated TF spectrogram. (b) Attention map output by DBA. (c) SSFJ-suppressed TF spectrogram. (d) SSFJ-suppressed SAR image. (e) SSFJ-contaminated SAR image.



(a) (b) (c) (d) (e)

Fig. 15. Jamming suppression results of SRSFJ + ISRJ. (a) TF spectrogram containing SRSFJ + ISRJ. (b) Attention map output by DBA. (c) TF spectrogram after suppressing SRSFJ + ISRJ. (d) SAR image after suppressing SRSFJ + ISRJ. (e) SAR image containing SRSFJ + ISRJ.

samples to discriminative features, and embeds all features into the same space. Thus, JRSNet can directly recognize jamming types not involved in training with high accuracy and generalization.

Jamming samples of types outside training datasets can be regarded as extremely imbalanced training samples. Therefore, the above recognition results within/outside training datasets show that under imbalanced training samples, the proposed feature distance metric idea and JRSNet have significant advantages.

F. Results of Jamming Suppression

We output the attention map of the first DBA in (13) and upsample it to the original TF spectrogram size. Then, 0.7 is used as a threshold for (10) to locate the jamming region, and the jamming is suppressed by notch filtering based on (11). Figs. 14(b) and 15(b) demonstrate the attention maps of SSFJ and SRSFJ + ISRJ. It can be seen that the complex jamming modulation information in TF spectrograms is precisely captured by DBA in JRSNet. The upsampling slightly enlarges the notch width in TF spectrograms, so the jamming suppression regions in Figs. 14(c) and 15(c) are larger than the actual, which results in a little loss of target information. Nevertheless, the suppression surface and multifalse targets in corresponding SAR images are successfully suppressed, as shown in Figs. 14(d) and 15(d), and the target scenes in SAR images become clear overall.

SAR images of different jamming signals are displayed in Fig. 16. Most jamming produces false targets of various shapes and positions, except for NAMJ and NAMJ + SSFJ which exhibit suppressive noise. Fig. 17 shows the SAR images after suppressing these jamming. Among them, ISFJ, ISRJ, CSJ, SRSFJ, and CSJ + ISFJ are effectively suppressed, and their imaging results are close to jamming-free SAR images. NAMJ and NAMJ + SSFJ overlap heavily with the target in TF spectrograms, and suppressing them leads to loss of target information, which brings distortion to SAR images, as shown in Fig. 17(f) and (h). This is a potential limitation of using JRSNet for jamming suppression. The quantitative indicators including SDR and SSIM values are put in Table X. Almost all SDRs are less than 0 and SSIMs are greater than 0.8, which implies good jamming suppression performance of JRSNet.

The above experimental results show that JRSNet can recognize and suppress multiple jamming signals synchronously, and it is of practical value.

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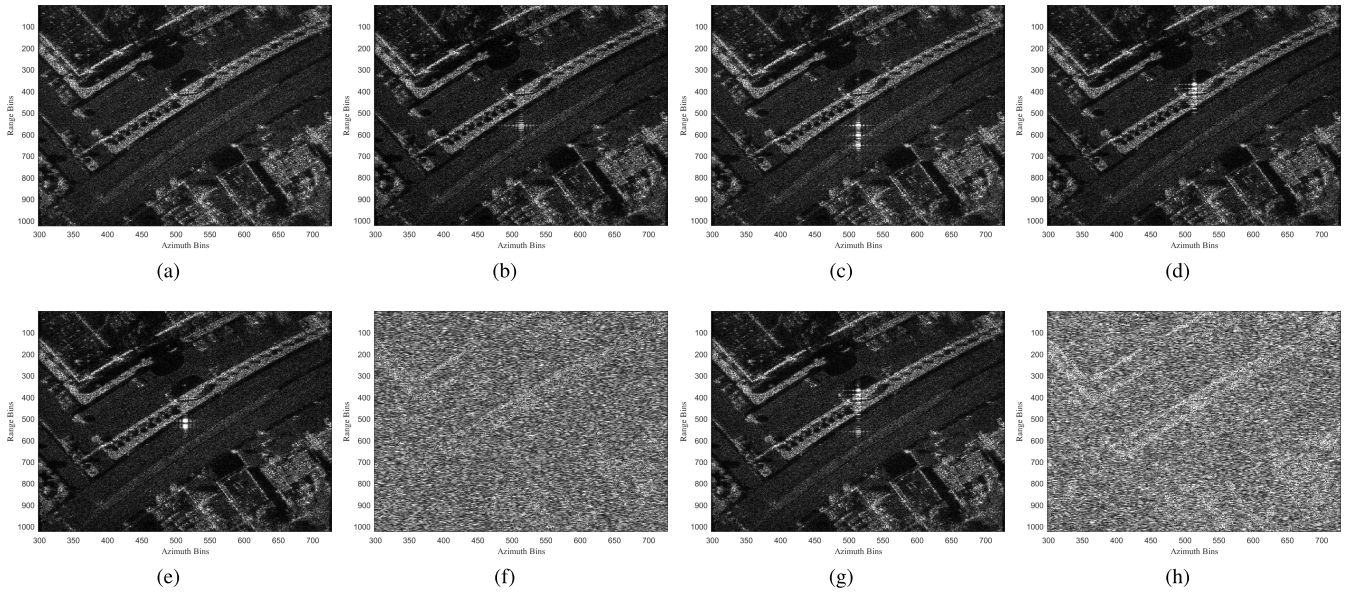


Fig. 16. SAR images with different jamming (except SSFJ and SRSFJ + ISRJ). (a) Jamming-free. (b) ISFJ. (c) ISRJ. (d) CSJ. (e) SRSFJ. (f) NAMJ. (g) CSJ + ISFJ. (h) NAMJ + SSFJ.

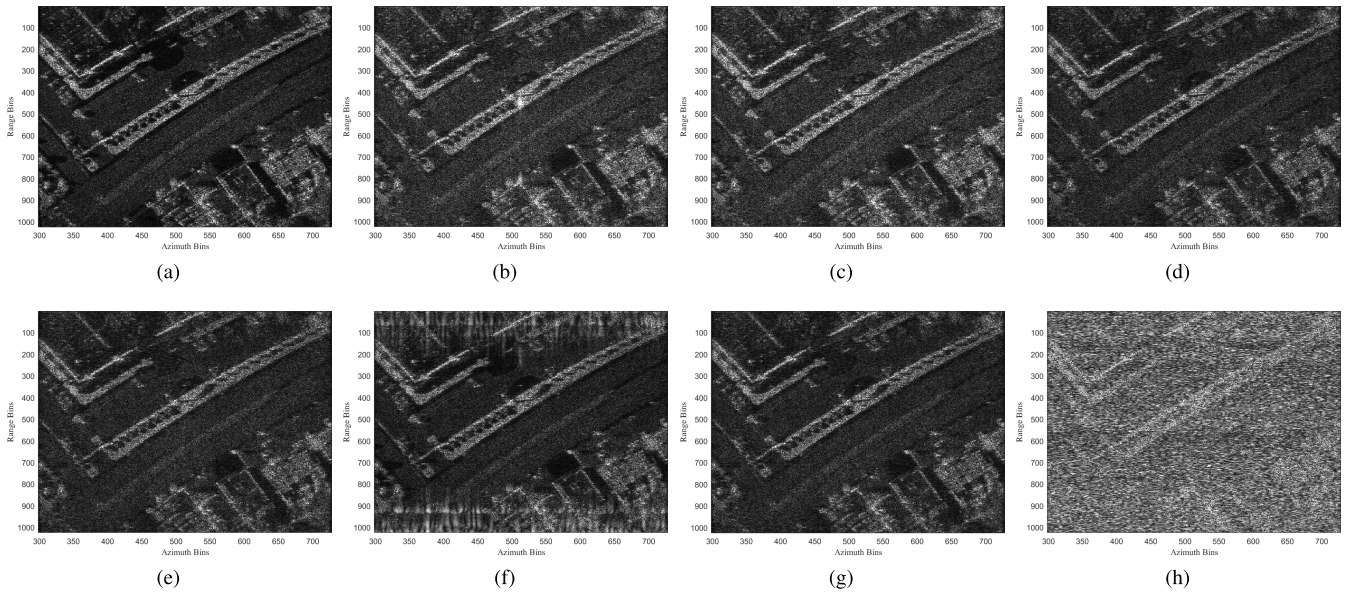


Fig. 17. SAR images after suppressing jamming (except SSFJ and SRSFJ + ISRJ). (a) Jamming-free. (b) ISFJ. (c) ISRJ. (d) CSJ. (e) SRSFJ. (f) NAMJ. (g) CSJ + ISFJ. (h) NAMJ + SSFJ.

TABLE X
SDR AND SSIM OF SAR IMAGES AFTER JAMMING SUPPRESSION BY JRSNET

Metric	ISFJ	ISRJ	CSJ	SSFJ	SRSFJ	NAMJ	SRSFJ+ISRJ	CSJ+ISFJ	NAMJ+SSFJ
SDR↓	-4.3088	-4.4450	-4.3666	-4.2739	-4.2654	-0.6748	-4.4345	-4.5396	8.6004
SSIM↑	0.8059	0.8077	0.8113	0.8008	0.8014	0.7201	0.8072	0.8267	0.3215

G. Results of Ablation

To determine the necessity of DBA, CAA, FD^{2}M loss, and DFCM, ablation experiments of the proposed JRSNet are carried out. Since DFCM is a component of FD^{2}M loss, the two are denoted together as FD^{2}M + DFCM. Table XI details the results of ablation experiments, including jamming recognition and suppression. From Table XI, compared to the baseline (trained by cross-entropy loss), the recognition accuracy and precision of FD^{2}M + DFCM are 5.11% and 7.41% higher, respectively. It demonstrates that under imbalanced training samples, FD^{2}M + DFCM makes JRSNet learn the discriminative feature expression paradigm for improving recognition performance. After incorporating DBA and CAA, the recognition performance is improved by an

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TABLE XI
RESULTS OF ABLATION EXPERIMENTS FOR JRSNET

Scheme			Jamming Recognition						Jamming Suppression	
DBA	CAA	FD^{2}M+DFCM	Accuracy(%)↑	Precision(%)↑	Recall(%)↑	F1 score(%)↑	Kappa(%)↑	Time(s)↓	SDR↓	SSIM↑
×	×	×	85.12±0.58	82.56±1.35	84.62±0.67	83.23±0.88	82.15±0.72	7.89±0.51	-	-
✓	×	×	85.96±0.88	84.12±0.72	85.11±1.08	83.92±0.76	83.21±0.55	8.86±0.24	-1.5632	0.4865
×	✓	×	86.03±1.23	83.46±1.34	84.91±0.57	83.81±0.98	83.08±0.46	8.31±0.30	-	-
×	×	✓	90.23±0.56	89.97±0.23	91.73±0.31	90.98±0.19	90.03±0.25	8.18±0.33	-	-
✓	✓	×	87.35±0.72	84.41±1.12	85.57±0.90	84.02±0.37	83.63±0.23	9.48±0.15	-2.3052	0.5006
✓	×	✓	92.64±0.21	93.15±0.50	92.71±0.69	92.09±0.87	91.14±0.82	9.28±0.43	-3.1868	0.6892
×	✓	✓	91.87±0.45	90.92±0.26	92.64±0.55	91.74±0.66	90.44±0.70	9.03±0.18	-	-
✓	✓	✓	93.21±0.35	94.08±0.19	93.96±0.22	93.87±0.44	91.82±0.57	9.40±0.46	-4.3002	0.8015

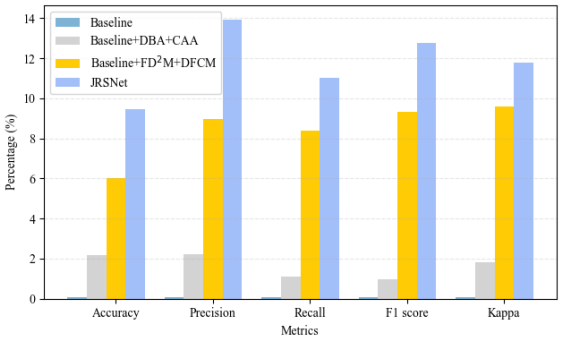


Fig. 18. Percentage growth of different jamming recognition evaluation metrics in ablation experiments.

average of 0.93% and 0.72%. Meanwhile, the jamming signals are initially suppressed by DBA. These verify that DBA and CAA can adaptively capture jamming regions in TF spectrograms, and then refine them to discriminative features. JRSNet integrates DBA, CAA, DFCM, and FD^{2}M loss, and achieves an optimal recognition accuracy of 93.21% and a maximum SSIM of 0.8015, comprehensively improving jamming recognition and suppression performance. Fig. 18 shows the recognition growth rate when different modules are added. DBA + CAA and FD^{2}M + DFCM increase the recognition growth rate by about 1.86% and 8.54%, respectively. The results further demonstrate the necessity of the proposed DBA, CAA, DFCM, and the FD^{2}M loss function.

VI. CONCLUSION

This article presents a novel discriminative feature distance metric idea and a specific model JRSNet for SAR jamming recognition under imbalanced training samples. Specifically, the DFCM and FD^{2}M loss guide JRSNet to learn an expression paradigm from jamming TF spectrograms to discriminative features, thus eliminating the negative effect of imbalanced training samples. Meanwhile, DBA and CAA capture jamming modulation differences from spatial and channel dimensions and then refine them into discriminative features. After training, the jamming type is predicted by the nearest distance metric of discriminative features. In addition, the jamming signal is synchronously suppressed by modulation information captured by the DBA. Visualization experiments verify the strong discriminability of features extracted by JRSNet. Compared with previous methods, JRSNet has the optimal recognition performance and effectively solves the imbalanced training samples issue. Furthermore, since JRSNet learns the jamming feature expression paradigm, it still maintains a strong generalization to jamming types uninvolved in training and has practical value. In future work, we will collect more measured SAR jamming samples, and further optimize JRSNet on this basis, so that it can recognize more jamming types, including azimuth direction, and implement real-time computation.

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